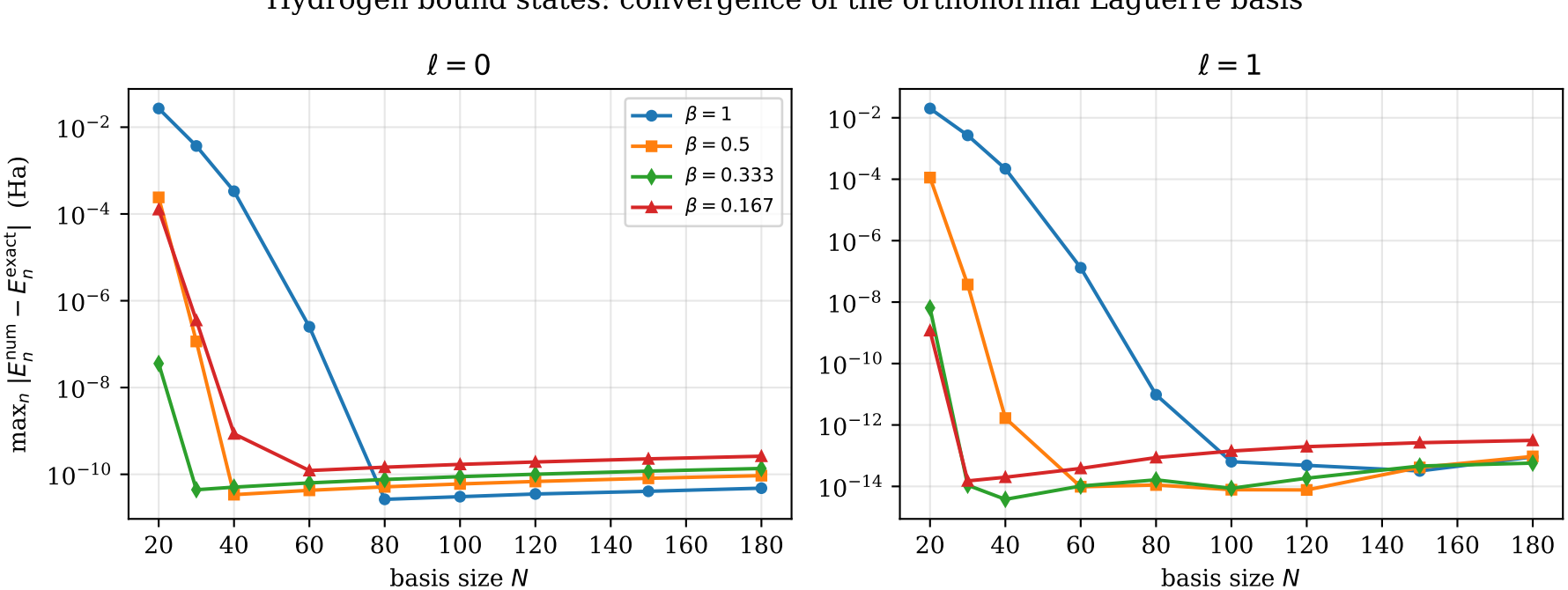
Hydrogen bound states: convergence of the orthonormal Laguerre basis
$\ell = 0$
$\ell = 1$
$\max_n |E_n^{\mathrm{num}} - E_n^{\mathrm{exact}}|$ (Ha)
basis size N
basis size N
$\beta = 1$
$\beta = 0.5$
$\beta = 0.333$
$\beta = 0.167$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
20
40
60
80
100
120
140
160
180